National University of Singapore
School of Computing
CS1101S: Programming Methodology
Semester I, 2026/2027

S10
CPS, Function Objects, and Loops

Problems:

Function Objects and State

1. Consider the following function `make_counter` and its use:

```
def make_counter():
    count = 0
    def next():
        nonlocal count
        count = count + 1
        return count
    return next

c1 = make_counter()
c2 = make_counter()
print(c1())
print(c1())
print(c2())
print(c1())
```

Draw the environment diagram after the last statement has been evaluated. What does each of the four calls of `c1` and `c2` return? Which frame is changed by each call? What would change if the declaration `nonlocal count` were removed from `next`?

CPS

2. The function `reduce` in Source Academy is giving rise to an iterative process and is implemented using CPS as follows:

```
def reduce(f, initial, xs):
    return _reduce(f, initial, xs, lambda x: x)

def _reduce(f, initial, xs, cont):
    return (cont(initial)
            if is_none(xs)
            else _reduce(f, initial, tail(xs),
                         lambda x: cont(f(head(xs), x))))

reduce(lambda x, y: x + y, 0, llist(1,2))
```

Generate the environment diagram for the `reduce` call above. Use the Source Academy to generate the diagram if you feel that it is too tedious to draw it by hand. Check in what order the 10 frames (excluding the global frame) get created. Which frames result from calls of continuations? In what order are the continuations invoked?

Loops and Functions

3. Consider the following program. What does it print? Explain!

```
xs_input = llist(1, 2, 3, 4)
output = None
i = xs_input
while not is_none(i):
    output = pair(lambda: i, output)
    i = tail(i)
print(head(tail(output))())
```